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Home advantage in tennis exists independent of competition level, gender and COVID-19 restrictions: evidence from German team tennis competitions

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ABSTRACT

While home advantage (HA) is a well-supported phenomenon in sports, the mechanisms responsible for its occurrence are not conclusively known. Research on HA in individual and amateur sport is scarce so far and situations where several plausible mechanisms do not apply are rare. Motivated by these aspects, the present study aims to investigate HA in an individual sport with a highly limited influence of three major factors (spectators, referees and travel) by investigating nearly 33,000 matches in almost 4000 tennis competitions from Germany. Based on separate analysis of competition levels (amateur to professional sports), competitions of both genders and competitions played before and during the COVID-19 pandemic, we find that HA exists in team tennis and its degree is not significantly influenced by these situational variables. These results support the notion that, in contrast to a widespread belief, an influence of spectators on athletes or referees is not a prerequisite for the existence of HA, at least in this sport. Moreover, HA seems to be a very general phenomenon existing widely independent of situational factors, suggesting that at least a significant part of HA is caused by physiological or psychological aspects like territoriality, expectations or learned behaviour.

HIGHLIGHTS

- Nearly 33,000 tennis matches in almost 4000 team competitions have been analysed.
- Strong evidence for the existence of home advantage in tennis independent of competition level, gender and COVID-19 restrictions.
- Results are evidence that home advantage exists in individual sports.
- Results suggest that spectator support or a spectator-induced referee bias are not required for home advantage.

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Home advantage; tennis; spectator influence; referee bias; gender

1. Introduction

Sports teams and partly also individual athletes have been found to perform more successful at home venues than away. This so-called home advantage (HA) is one of the best empirically supported phenomena in the domain of sports. However, it remains

less clear which mechanisms are responsible for its occurrence (Wunderlich, Furley, & Memmert, 2021). Most likely the HA is a multi-causal phenomenon that is explained by several factors and their interplay. However, it has proven difficult to provide converging evidence for the theorised mechanisms contributing to the HA because researchers cannot actively change focal variables in real-world sport events to measure the impact on the degree of HA as this is typically done in experimental research to determine causality. On the other hand, experimental research on the HA that could help illuminate the causes of the HA faces the challenge of ecological validity as it typically cannot reproduce the entire complexity of a real-world sport event. To account for these difficulties, a fruitful avenue of research has investigated the effect of naturally occurring contextual factors like the presence of a supporting crowd on the magnitude of the HA. The present research attempts to add to this literature by investigating the HA in tennis as a function of competition level, gender and COVID-19 restrictions. Given the characteristics of tennis in general and these three aspects in particular in relation to theorised mechanisms like referee bias, travel distance or territoriality, our data provide an exceptional empirical test of theorised mechanisms of the HA.

Several plausible mechanisms have been discussed as potential contributors to the HA (Allen & Jones, 2014; Courneya & Carron, 1992; Nevill & Holder, 1999; Pollard, 2008) or possible confounding factors (see Riedl et al., 2014). Two main situational factors contributing to the HA (see Allen & Jones, 2014 for a review) are assumed to be the support of the home audience and referee decisions favouring the home team. There is empirical support for other situational factors like travel fatigue of the away team (Goumas, 2014b; Recht et al., 1995), however, such effects appear to be decreasing over time (Beckmann, 2021). For other factors like facility familiarity only little support can be found in empirical data (Loughead et al., 2003). So arguably the strongest empirical support for factors leading to the HA are spectator presence and referee biases giving an advantage to the home team and a disadvantage to the away team (for a review Gómez-Ruano et al., 2021). In support of audience effects, studies have demonstrated that components of spectators/fans (e.g., size, density, and proximity) show a correlation with the magnitude of the HA (e.g., Armatas & Pollard, 2014; Goumas, 2014b, 2014c; Nevill et al., 1996) and the spectators themselves believe that their behaviour (such as booing, fighting and cheering) contributes to the HA (Wolfson et al., 2005). While the exact mechanisms of how supportive audiences might impact athletes (e.g., via a motivational boost for the home team or increased anxiety for the away team) remain unclear and direct effects of spectators on performance are discussed controversially (see Strauss, 2002; Strauss et al., 2021), there is solid evidence of audience effects on official's decision making (Downward & Jones, 2007; Goumas, 2014a; Unkelbach & Memmert, 2010). Unequal treatment of home and away teams can be found in match statistics in soccer with regard to both injury time (Garicano et al., 2005; Riedl et al., 2015; Scoppa, 2008) and disciplinary sanctions (Goumas, 2014a; Sutter & Kocher, 2004). Moreover, it has been shown that these differences actually depend on the presence of a crowd (Pettersson-Lidbom & Priks, 2010) and largely disappear in matches without spectators (Scoppa, 2021; Wunderlich et al., 2021). These differences are commonly assumed to be caused by the referees consciously or unconsciously favouring the home team as a reaction to social pressure (Garicano et al., 2005; Scoppa, 2008). Experimental studies support this reasoning as they demonstrate that referees are affected by crowd noise and tend to decide more favourably for the home team as

opposed to the away team (Nevill et al., 2002; Unkelbach & Memmert, 2010). This officiating bias might be the result of various factors (Allen & Jones, 2014) like motivation (i.e., referees are reluctant to displease the crowd) or that crowd noise serves as decision heuristic and thereby implicitly biases referees towards the opinion of the dominant crowd.

A less situational and more general theoretical model assumes the HA to be a manifestation of a natural protective physiological and behavioural responses to protect one's perceived territory (Neave & Wolfson, 2003). This line of reasoning builds on evolutionary theorising (Furley, 2019; Neave & Wolfson, 2003) by arguing that better performance in what athletes perceive as their home territory can be considered a manifestation of the natural protective response to territorial incursion. Allen and Jones (2014) assume that the territoriality response (including higher levels of testosterone) contributes to the HA by increasing risk-taking behaviour, improving metabolic rates of muscles, and by improving spatial ability (M. Jones et al., 2005). A further mechanism as to how the territoriality response might contribute to the HA is highlighted by evidence showing that athletes seem to change their body language depending on game location (Furley et al., 2018). These changes in body language have been argued to potentially impact the confidence levels of both the athletes displaying a certain body language and the opponents observing the body language (Furley & Dicks, 2012; Furley, Dicks, & Memmert, 2012; Furley, Dicks, Stendtkke, et al., 2012; Furley & Schweizer, 2014, 2020). Additional research supports the notion that confidence levels of athletes play an important role in explaining the HA (Waters & Lovell, 2002).

Driven by the question of whether situational effects or more general physiological and psychological aspects are responsible for the HA, we attempted to test the HA in a natural context that excludes, or at least strongly reduces, the influence of several situational factors: tennis league games. In contrast to other sports, it is absolutely unusual to cheer or even disrupt players while a point is in progress in professional tennis. While players might be supported by a home crowd between points, the degree of emotional reactions of tennis spectators are usually not comparable to those of, for example, soccer fans. Moreover, referee bias appears to be widely irrelevant in tennis, as referees are hardly in need to take controversial and subjective decisions. This is particularly true if, like in the present analysis, matches are played on clay court where the ball leaves a mark when touching the ground and close calls can be assessed very objectively even after a point. We would like to acknowledge that tennis matches are not entirely free of subjective referee decisions, for example in the case of code violations (like racket abuse). However, such decisions seem to be quite rare and have a limited influence on the final result, particularly compared to sports like soccer, basketball or handball with a high density of referee decisions that contain a subjective component. Likewise, the inclusion of amateur competitions offers an interesting context because the number of spectators in amateur matches is negligible and amateur competitions are usually organised with a very regional character. Thus, two potentially relevant factors contributing to HA, namely spectator influence and travel distances, can be assumed to play a highly limited role, if at all. The same is true for referee bias, as amateur competitions in tennis are played in absence of a referee, i.e., in contrast to other sports the players themselves take all relevant decisions. The theoretical role of familiarity in tennis is less clear, as team tennis competitions are not comparable to inherent team sports and players can additionally or predominantly compete in individual tournaments. Professional "players

travel from tournament to tournament” (Koning, 2011, p. 20), which means that they commonly travel from a tournament to their “home” matches, for a highly limited amount of team matches per season. In terms of facility familiarity, this is not comparable to players in other team sports without individual tournaments, who play regularly in the same home facility for a large amount of matchdays per season. Thus, it seems rather unlikely that a professional tennis player predominantly competing in national or international tournaments benefits from an increased familiarity or territoriality if playing at home. Amateur players, however, having a long-term loyalty and constantly training in the own club’s facility, might draw a greater benefit from playing at home. As clay is the standard surface for outdoor matches played in Germany, facility familiarity does not refer to the type of surface such as carpet, hard court, grass and clay (see Koning, 2011). However, even for clay courts, small differences in the amount of clay and specific bouncing behaviour of the ball might add to the facility familiarity of the home players.

An additional motivation for investigating tennis is its character as an individual sport. While there is a large body of literature covering evidence for the HA in team sports (e.g., baseball (M. B. Jones, 2015); basketball (Gómez & Pollard, 2011); soccer (Pollard & Gomez, 2014); ice hockey (Leard & Doyle, 2011)), analyses on HA in individual sport are relatively rare and the results are less clear (Giménez-Egido & Torres-Luque, 2021; M. B. Jones, 2013; Koning, 2005, 2011). The difficulty of analysing HA in individual sports can partly be explained by the fact that individual athletes might take part in competitions in their own country, but usually do not have one clearly assigned home venue where they compete regularly in a balanced competition schedule. Tennis offers a particularly favourable opportunity for straightforward analysis of HA in an individual sport, as there are tennis competitions where individual matches are embedded into team formats with clearly assigned home venues.

Existing literature on the HA in tennis and amateur sports is scarce. With regard to HA in professional tennis the results are mixed and partly even contradictory. It needs to be acknowledged that the investigation of HA in tennis tournaments is complicated by the aforementioned absence of balanced schedules and thus models need to control carefully for factors like playing strength or playing surface (Holder & Nevill, 1997; Koning, 2011; Nevill et al., 1997). Results of Grand Slam tournaments provided little evidence in favour of HA (Holder & Nevill, 1997; Nevill et al., 1997). Inconsistent with this, significant HA was found in a dataset of ATP tennis tournaments (Koning, 2011) as well as in the Davis Cup (Maquiriain & Baglione, 2017). Existing results suggest that HA tends to be lower in women’s tennis compared to men’s tennis as for both individual tournaments (Koning, 2011) as well as Davis Cup and Fed Cup matches (Maquiriain & Baglione, 2017) significant HA for men and only non-significant HA for women was reported. These differences, however, need to be treated with some caution and are partly explainable by lower sample sizes used in the analysis of women’s competitions. Moreover, both for male (Davis Cup) and female (Fed Cup) team competitions, evidence for a decrease or even reversion of HA in situations of maximum pressure have been reported (Gayton et al., 2013, 2009). There is scarce literature suggesting that the HA exists in amateur soccer, but inconsistent conclusions about differences between professional and amateur leagues have been reported (Almeida & Volossovitch, 2017; Armatas et al., 2013; Wunderlich et al., 2021). All in all, the HA in amateur sports seems to be highly understudied, particularly considering that large quantities of data on the results of

amateur competitions are available. Given the importance of furthering the understanding of the theoretical mechanisms leading to the HA, it is surprising that the context of team tennis and in particular amateur tennis have not been studied to date.

In summary, the purpose of the present study is to investigate the existence of HA in an individual sport with an arguably highly limited influence of three major factors (supporting crowds, referee, and travel). We do this by drawing on an extensive self-generated dataset including nearly 33,000 individual matches from almost 4,000 team competitions including both professional and recreational tennis players. Moreover, by using information on three situational factors, we shed light on the influence of competition level, gender and COVID-19 restrictions on the HA. The present study has the potential to make substantial contributions to the literature on the HA. First, it adds to the limited knowledge so far on the HA in individual sports and amateur sports. Second, to the best of our knowledge, except for national teams in Davis Cup or Fed Cup matches (Gayton et al., 2009, 2013; Maquiriain & Baglione, 2017), the HA in team tennis has never been investigated before. Third, and most importantly, testing the existence of the HA in tennis league games across different performance levels, genders and COVID-19 restrictions provides an exceptional test of recent theorising on the HA in sports as the existence of HA in this specific context, could be considered as further evidence for the idea of HA being driven by less situational, but more general mechanisms including the territoriality hypothesis. The present research tests one main hypothesis and three side hypotheses. As discussed before, tennis in general and amateur tennis in particular should be less affected by spectator support, referee bias or travel distances. However, there are still theoretical reasons to assume that home teams can have an advantage, particularly due to territoriality (Neave & Wolfson, 2003), effects of expectation (Waters & Lovell, 2002) and learning (Staufenbiel et al., 2018) or partly familiarity (Loughead et al., 2003). Thus, our main hypothesis is that *the HA exists in tennis*. In addition to our main hypothesis we further explore potential differences in the HA as a function of competition level, gender and COVID-19 restrictions. Inconsistent results on differences between HA in amateur and professional leagues (Almeida & Volossovitch, 2017; Armatas et al., 2013) drive the question whether HA is generally affected by the level of competition. Driven by theoretical differences particularly with regard to familiarity and spectators, we hypothesise that *competition level has an influence on the degree of the HA in tennis*. The HA has been predominantly investigated in men's sports and has hardly been studied in women's sport. The scarce results on HA in women's sports are mixed, but the effect of HA in women's competitions, if existent, tends to be smaller than in men's competitions (Gayton et al., 1987; Koning, 2011; Pollard & Gómez, 2014). Moreover, if testosterone, as suggested by Neave and Wolfson (2003) actually plays a role in the emergence of HA, this would be a potential, but not yet evidenced, driver for differences between HA in men's and women's competitions. We hypothesise that, in line with this theoretical consideration and the scarce empirical results, *gender has an influence on the degree of HA in tennis*. Finally, spectator exclusion in sports as a measure to contain the COVID-19 pandemic has stimulated a lot of research using this natural experiment to analyse the influence of spectators on HA. Results show that HA tended to decrease, but still exists in team sports during the COVID-19 pandemic (Bryson et al., 2021; Higgs & Stavness, 2021; McCarrick et al., 2021; Scoppa, 2021; Sors et al., 2021; Wunderlich et al., 2021). However, given that the influence of spectators on

players and referees in (amateur) tennis can be considered to be relatively small, we hypothesise that in contrast to results from other sports, *COVID-19 restrictions have no influence on the degree of HA in tennis*.

2. Methods

2.1. Competition format

The present analysis focuses on official team tennis competitions played in league formats throughout Germany. No additional tournaments with individual players – as usual in professional tennis at an ATP or WTA level – have been considered. Teams are divided into groups, where the same league level can consist of several groups. Within these groups, all teams play in a single round-robin format, i.e., there is only one match against each other and not one home and one away match each as common in a double-round-robin format in most team sports. For reasons of fairness, it is ensured that all teams have approximately the same number of home and away matches. Within a team competition, each player competes in one singles match as well as in one doubles match. Each team, depending on the level of competition, consist of 4 or 6 players, which results in a total of 6 or 9 matches per competition. A match follows the common rules of a tennis match (see O'Malley, 2008). All matches in our database are played in the best-of-three format, where the third set is replaced by a so-called champions tie-break (tie-break played until one competitor reaches 10 points with at least a two-point margin). For each match won, a team is awarded one point and at the end of the team competition the team that has scored the most points is declared the winner. If the teams have scored the same number of match points, the team competition is considered to be a draw. For reasons of clarity, we refer to *competitions* when assessing the winner of a team competition and to *matches* when assessing the winner of an individual match in the context of a team competition.

2.2. Data and situational variables

The dataset consists of tennis team competitions from Germany including four consecutive seasons (2018–2021). For each team competition, information on date, gender, competition level, points (i.e., matches won) of home and away team as well as the winner of the competition (home, draw away) were collected. Only regular-season games are considered as promotion and relegation games are an exception taking place only in specific leagues and situations. The sample size of such specific matches with extraordinary mental conditions thus appears too small to obtain meaningful results. The team competitions are split into a summer season played outdoor and a winter season played indoor. We only considered the outdoor season and excluded the indoor season for two reasons. First, the competitions with the highest competition levels only take place in the summer season. Second, while clay is the standard surface for outdoor matches, there is higher heterogeneity of surfaces for indoor matches including carpet, hard court or clay. Scheduled competitions that were cancelled for some reasons are excluded from the dataset. Moreover, competitions with a drawn result were excluded from analysis of competitions

won, but included in analysis of individual matches won. This results in a total dataset of 3815 competitions and 32,998 matches.

For analysis, the dataset was split by three situational factors. First, the competition level ranging from professional players to recreational players in a total of eleven different league levels that are grouped into three categories. We refer to *professional* competitions for the highest two cross-regional leagues 1. *Bundesliga* and 2. *Bundesliga* (529 competitions; 4673 matches). We refer to *semi-professional* competitions for the two regional leagues *Regionalliga* and *Oberliga* that can be considered to be in-between professional and amateur playing level (1258 competitions; 11,313 matches). All leagues from 1. *Verbandsliga* downwards are considered to represent full amateur structures and are denoted as *amateur* competitions (2028 competitions; 17,012 matches). The data for the highest three leagues cover all competitions across Germany, while we restricted the data retrieval for all lower leagues to one regional tennis association (Tennisverband Mittelrhein) and thus excluded the other regional associations, due to the high quantity of competitions in lower leagues. Data for the professional competitions were obtained from the official website of the German tennis federation (Deutscher Tennis Bund, 2021), data for the third league (Regionalliga) were obtained from the websites of the four regional leagues (Tennis Regionalliga Nord-Ost, 2021; Tennis Regionalliga Süd-Ost, 2021; Tennis Regionalliga Süd-West, 2021; Tennis Regionalliga West, 2021) and data from the fourth league (Oberliga) downwards were obtained from the website of one regional tennis association (Tennisverband Mittelrhein, 2021). Second, the gender including both women's (1531 competitions; 13,346 matches) and men's competitions (2284 competitions; 19,652 matches). Third, the COVID-19 pandemic, which led to different and varying restrictions regarding societal restrictions, hygienic measures and spectator involvement. Data were split for seasons 2018 and 2019 taking place under normal circumstances *before the pandemic* (2219 competitions; 19,267 matches) and seasons 2020 and 2021 taking place *during the pandemic* (1596 competitions; 13,731 matches). Please refer to Table 1 for further details on the dataset and the competition levels involved.

2.3. Statistical analysis

In the context of the present study, statistical tests are needed to answer two different sorts of questions: First, does the HA exist for certain situations, i.e., is the home team significantly more successful than the away team? To do this we directly compare the number of competitions/matches won by the home and the away teams. Second, do certain situational factors influence the HA, i.e., are there differences in the degree of HA if splitting the data for certain situational factors? To do this we compare the ratio between home and away wins/matches across different situations. For both types of questions, the results of χ^2 -tests are reported. We consider an effect to be significant if the p -value falls below 0.05 and we distinguish significance levels of 0.05, 0.01 and 0.001 in the detailed results. We additionally report effect sizes (Cohen's w) to quantify the degree of home advantage. However, please note that in the present study home advantage is tested by comparing home wins/matches to away wins/matches with a χ^2 -goodness-of-fit-test and in this specific case Cohen's w is simply equal to the relative difference between home wins/matches and away wins/matches (e.g., Cohen's w would

Table 1. Information on the data set used within this study.

Level of play	League	Number of team competitions excluding draws	Number of matches within the team competitions
Professional	1. Bundesliga	163	1379
	2. Bundesliga	366	3294
Semi professional	Regionalliga	1076	9684
	Oberliga	182	1629
Amateur	1. Verbandsliga	437	3876
	2. Verbandsliga	662	5915
	1. Bezirksliga	258	2184
	2. Bezirksliga	228	1866
	1. Kreisliga	127	1001
	2. Kreisliga	140	995
	3. Kreisliga	176	1175
	Total	3815	32,998

be 0.10 = 55% – 45% in case of 55% home wins and 45% away wins). As such, the percentage of home wins is already a measure of the home advantage that can be reasonably compared across various situations or to the results of comparable studies. All calculations have been performed using the R software environment for statistical computing (R Core Team, 2021).

3. Results

3.1. Overall

Based on the entire dataset, i.e., across all situational factors, home teams win around 53% of all matches and competitions. For both measures, this represents highly significant ($p < .001$) evidence for the existence of a HA in team tennis in line with our main hypothesis (see Table 2).

3.2. Level of competition

Table 3 summarises the results with reference to different competition levels. The relative number of home competitions won range from 52.9% to 53.9% and the relative number of home matches won range from 52.3% to 52.9%, both being highly homogenous across all groups. Except for the number of competitions in professional leagues, this represents significant evidence for the existence of a HA across all groups for both measures (competitions and matches). Given the percentage of professional home competitions won indicating an even slightly stronger effect than for semi professional and amateur

Table 2. Overall results on the home advantage.

Competitions	Home	Away	Home in %	Away in %	χ^2	df	p -value	Cohen's w
All competitions	2030	1785	53.2	46.8	15.73	1	<.001*	0.064
Matches	Home	Away	Home in %	Away in %	χ^2	df	p -value	Cohen's w
All competitions	17,381	15,617	52.7	47.3	94.30	1	<.001*	0.053

Notes: The table shows the absolute and relative number of competitions (matches) won at home and away as well as results of χ^2 -tests for the existence of home advantage including effect sizes. Significant results are marked with an asterisk.

Table 3. Home advantage across different competition levels.

Competitions	Home	Away	Home in %	Away in %	χ^2	df	<i>p</i> -value	Cohen's <i>w</i>
Professional	285	244	53.9	46.1	3.18	1	.075	0.078
Semi professional	665	593	52.9	47.1	4.12	1	<.05*	0.057
Amateur	1080	948	53.3	46.7	8.59	1	<.01*	0.065
Matches	Home	Away	Home in %	Away in %	χ^2	df	<i>p</i> -value	Cohen's <i>w</i>
Professional	2442	2231	52.3	47.7	9.53	1	<.01*	0.045
Semi professional	5946	5367	52.6	47.4	29.63	1	<.001*	0.051
Amateur	8993	8019	52.9	47.1	55.77	1	<.001*	0.057

Notes: The table shows the absolute and relative number of competitions (matches) won at home and away as well as results of χ^2 -tests for the existence of home advantage distinguishing between three different levels of competition (professional, semi professional, amateur) including effect sizes. Significant results are marked with an asterisk.

competitions as well as the significant HA for professional matches, this failure to reach significance appears to be no peculiarity of professional competitions, but a direct consequence of the relatively more limited sample size of 529 competitions. χ^2 -tests revealed no significant results if testing for differences in HA for the three different levels of competition both with regard to competitions ($\chi^2 = 0.16$, $df = 2$, $p = 0.925$) and matches ($\chi^2 = 0.63$, $df = 2$, $p = 0.731$). Thus, no significant influence of the competition level on HA was found.

3.3. Gender

Considering the results of the gender comparison summarised in Table 4, home win percentages again are highly homogenous and represent an increased winning percentage for home teams. Thus, a significant HA can be observed for both women and men with reference to both measures. While the degree of HA appears to be minimally smaller for female players, this difference is not sufficient to confirm a significant difference between the genders with regard to the number of competitions ($\chi^2 = 0.04$, $df = 1$, $p = 0.834$) as well as the number of matches ($\chi^2 = 1.27$, $df = 1$, $p = 0.259$). Thus, no evidence for gender differences in the HA can be attested.

3.4. COVID-19

Both in the 2018/2019 summer seasons, which took place under “normal” conditions before the pandemic and in the 2020/2021 summer seasons, which suffered the effects of the COVID-19 pandemic, a significant HA has been found ($p < .01$ each with regard

Table 4. Home advantage for male and female competitions.

Competitions	Home	Away	Home in %	Away in %	χ^2	df	<i>p</i> -value	Cohen's <i>w</i>
Male	1219	1065	53.4	46.6	10.38	1	<.01*	0.067
Female	811	720	53.0	47.0	5.41	1	<.05*	0.059
Matches	Home	Away	Home in %	Away in %	χ^2	df	<i>p</i> -value	Cohen's <i>w</i>
Male	10,402	9250	52.9	47.1	67.53	1	<.001*	0.059
Female	6979	6367	52.3	47.7	28.06	1	<.001*	0.046

Notes: The table shows the absolute and relative number of competitions (matches) won at home and away as well as results of χ^2 -tests for the existence of home advantage distinguishing between male and female competitions including effect sizes. Significant results are marked with an asterisk.

Table 5. Home advantage for competitions before and during the COVID-19 pandemic.

Competitions	Home	Away	Home in %	Away in %	χ^2	df	<i>p</i> -value	Cohen's <i>w</i>
Before pandemic	1173	1046	52.9	47.1	7.27	1	<.01*	0.057
During pandemic	857	739	53.7	46.3	8.72	1	<.01*	0.074
Matches	Home	Away	Home in %	Away in %	χ^2	df	<i>p</i> -value	Cohen's <i>w</i>
Before pandemic	10,103	9164	52.4	47.6	45.76	1	<.001*	0.049
During pandemic	7278	6453	53.0	47.0	49.57	1	<.001*	0.060

Notes: The table shows the absolute and relative number of competitions (matches) won at home and away as well as results of χ^2 -tests for the existence of home advantage distinguishing between competitions before and during the COVID-19 pandemic including effect sizes. Significant results are marked with an asterisk.

to competitions; $p < .001$ each with regard to matches). HA even increased very slightly during the pandemic, but again this difference clearly failed to reach significance with regard to competitions ($\chi^2 = 0.23$, $df = 1$, $p = 0.633$) and matches ($\chi^2 = 1.01$, $df = 1$, $p = 0.314$). Thus, no significant influence of COVID-19 restrictions on the HA exists (Table 5).

4. Discussion

The goal of the present research was to investigate the HA in league tennis and thereby challenge contemporary theorising that has argued that the most important factors contributing to the HA advantage are supporting spectators at home and biased referee decisions. In this respect, the present study provides strong evidence for the existence of the HA in team tennis without supporting home crowds and referees. This finding also shows that the phenomenon of the HA is transferrable from team sports to individual sports or at least to an individual sport embedded in a team format, being in contrast to the argument of M. B. Jones (2013) that evidence for HA in individual sports is weak and even in team sports its degree decreases if the extent of teamwork is limited (M. B. Jones, 2015). Given the special characteristics of (amateur) tennis, the factors actually driving this HA are not clear at all and need careful discussion. The highly renowned work of Courneya and Carron (1992) states familiarity with the venue, spectators, travel stress, and rules as four possible factors contributing to HA. The home setting, which offers the players protection and is usually associated with positive training experiences or family experiences, could therefore be seen as a reason why athletes perform better. While this is particularly true in amateur tennis, it seems highly questionable whether a professional tennis player playing in international tournaments who plays for his home club actually feels more familiar with the home courts and surrounding than his opponent being in a comparable situation, but contesting for the away team. Spectators, if at all, are highly limited in size in amateur tennis and in crowd noise during points in professional tennis. As such, spectators, from a theoretical standpoint should not be a significant factor in the present dataset. The same is true for travelling distances, as it can be assumed that in the case of amateur games in regional leagues, the distances are not great enough to influence performance and game results. While distances are higher in nationwide games in the highest leagues, professional players travel from tournament to tournament anyways interrupted by team matches and thus the travel burden for the home team should be almost comparable to that of the away team. Moreover, there exists no rule advantage for the home team in professional or amateur team tennis that might have the potential to significantly affect results. Another highly discussed factor in the literature is a (crowd-

induced) referee bias (Constantinou et al., 2014; Nevill et al., 2002; Unkelbach & Memmert, 2010; Wunderlich et al., 2021), which, however, likewise appears neglectable in tennis due to the limited spectator size and noise as well as the lack of subjective decisions in professional tennis and the total lack of referees in amateur tennis. This leaves physiological and psychological factors as potential explanation for the HA found, such as territoriality (Neave & Wolfson, 2003), expectations (Waters & Lovell, 2002) and learning (Staufenbiel et al., 2018).

The characteristics of the competitions are highly heterogeneous across different situations in this study, for example with reference to player quality, importance of matches, referee involvement, crowd size, regionality, involvement of international players, etc. In this context, it can be considered surprising that the HA in team tennis has been shown to be strikingly homogenous. Considering all groups, the home players winning percentage on a match level range from 52.3% to 53.0% and neither the competition level, nor the gender or the COVID-19 restrictions significantly affected the degree of HA. This supports the idea of the HA as a very general phenomenon, potentially driven by an evolved territoriality response. According to this theorising, the HA is considered to be a manifestation of a natural protective physiological and behavioural responses to protect one's perceived territory (Neave & Wolfson, 2003). More specifically, better athletic performance is considered to be a by-product that occurs via a cascade of physiological and behavioural reactions as a consequence of territorial incursion in what athlete's consider to be their home territory. However, as we only analysed winning percentages and none of the assumed mediating variables like hormone concentrations or body language measures, this interpretation remains highly speculative. In order to further support this assumption, future research should focus on measuring testosterone and cortisol levels as well as mental self-report measures. Such studies on physiological and psychological states of players have already been conducted in team sports such as ice hockey and, accordingly, could be replicated in tennis (Carré et al., 2006).

While the present data reveal clear results with regard to significance, these might partly be a consequence of the relatively large sample sizes and do not readily imply substantial or practically relevant effect sizes. For this reason, we would like to put the results into the context of HA effects found in other sports. For comparison, we use the average HA across several European leagues in women's and men's soccer reported by Pollard and Gómez (2014) as well as the HA reported for several North American team sports with respect to the last season included in the long-term analysis of Pollard and Pollard (2005). Given the overall HA of 53.2% for tennis competitions, the aforementioned studies have reported comparable degrees of HA for ice hockey (NHL, 54.0%), baseball (AL, 53.7%; NL, 54.5%) and women's soccer (26 European leagues, 54.2%), but higher degrees of HA for American soccer (NFL, 58.0%), basketball (NBA, 62.0%) and men's soccer (26 European leagues, 60.0%). As such, home advantage in tennis appears to be comparable to team sports at the lower spectrum of HA, which would be in line with the idea of HA as a multi-causal phenomenon, where only some of the theorised mechanisms apply to the tennis competitions in our dataset, while other sports and leagues are more exposed to further mechanisms like travel distances (e.g., North American leagues) or referee bias (e.g., basketball or men's soccer as sports with high degree of subjectivity in referee decisions and a high number of spectators).

Although a minimally smaller degree of HA was found for women's matches, no significant difference between the genders was found in our database and a significant HA also exists for women. From a theoretical standpoint, there would be mechanisms that could explain differences. For example, possible different hormonal reactions of female and male players to defend own territory in line with the idea of territoriality. Our results suggest that either no such differences exist (speaking against the territoriality reasoning) or that existing differences do not affect HA. Moreover, situational factors might drive gender-induced differences. Most prominently a higher number of spectators, but also higher financial incentives and public interest, particularly present for male team sports. In line with this reasoning, it seems possible that HA can be decreased in some female (team) sports (see Pollard & Gómez, 2014), but not in general.

The results might also be explained (or partly explained) by several other factors. For example, our results suggest that the degree of HA is independent of the competition level, which might be considered somewhat surprising, as we would argue that familiarity and emotional connection to their club should be smaller for professionals than for amateurs, which, however, is not reflected in the results. A possible explanation for this, that we cannot fully rule out with the current dataset, is that different opposing effects on HA balance each other out. For example, professional players might benefit from increased effort of the home clubs such as offering higher financial winning bonus or a certain motivational effect of spectator support, which is not the case in amateur matches. At the same time, amateur players might, in contrast to the professional players, benefit from increased familiarity with their home clubs' courts. Other potential factors are expectations and learned behaviour, that are widely independent of the exact circumstances of a competition. For example, the HA could be partly conditioned by the self-fulfilling prophecy, where the inner attitude and the belief that they can perform better at home implies that the tennis players feel strengthened in their abilities and show increased physical or mental strength against their opponents.

Finally, we also did not find significant result with regard to the distinction of HA before and during the COVID-19 pandemic. During the 2020 and 2021 seasons, the home win percentage (53.7%) was even slightly higher compared to the 2018 and 2019 seasons (52.9%) under regular conditions, which again does not represent a significant difference. These results need to be treated with some caution, as the COVID-19 measures during the pandemic changed regularly and were not uniform in different regions. As such, the distinction between both groups of competitions may still involve a variety of different situations. The large body of literature on COVID-19-induced changes to the HA mainly focuses on the influence of spectators (Bryson et al., 2021; Higgs & Stavness, 2021; McCarrick et al., 2021; Scoppa, 2021; Sors et al., 2021; Wunderlich et al., 2021). With regard to this aspect, our data clearly suggests that the presence of social interaction in general and spectators in particular is no prerequisite for HA (in tennis). The focus on spectators in the pandemic is a bit one-dimensional as COVID-19 measures might have had significant other effects on training volume and physical fitness as well as general wellbeing or mental problems. Unfortunately, we cannot investigate potential influence of these aspects on the degree of HA, due to the difficulty to collect such data.

The present study is subject to some limitations that partly are related to the dataset used and could be a fruitful field of further research in (amateur) tennis. While our dataset is large in quantity, it is limited in granularity. More detailed match results, i.e., on the exact outcomes of sets might reveal motivational differences like decreased effort after losing the first set in away matches. Separate information for the number of singles and doubles matches might enable insights on whether HA is more pronounced in a “team” of two players. Exact spectator numbers might reveal whether HA is to some extent influenced by the crowd size in professional tennis. Moreover, analysis of player strength in professional matches would help to reveal whether teams chose a more promising lineup in home matches. By including youth competitions or senior competitions it would be possible to understand whether HA exists equally across all groups of ages. Finally, team tennis competitions from other countries might reveal influence of cultural differences on HA. However, not all of these information are publicly available and obtaining these data is difficult. We would also like to acknowledge that in tennis the player skills can be highly heterogenous at the same league level. As players are ranked from one to six (or one to four) by playing strength in each competition, it is possible that the variability within one league can be comparable or even higher than the variability across leagues and as such the level of competition involves a certain degree of imprecision.

5. Conclusions

In conclusion, the present data from tennis are evidence that the HA exists in an individual sport, that is not highly dependent on subjective decisions neither with regard to referees judging the rules nor judges evaluating the execution of movements. Moreover, they strongly suggest that HA does not require spectators, spectator-induced referee bias or significant travel distances, does not require a high level of competition, exists for both genders and that even the degree of HA is highly homogeneous across many different situations.

As such, it appears that at least a significant part of the phenomenon of the HA is related to general physiological and psychological aspects and not dependent on situational factors. For this reason, coaches might consider psychological coaching and mental training in an attempt to develop strategies to adjust expectations and mental states in away matches. To gain more direct evidence on the influence of these aspects of the HA, studies involving measurements of testosterone and cortisol levels and mental self-report measures could establish a clearer link between physiological and psychological states and competition performance at home and away.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

All data used within this study are available in the public domain and all data sources have been cited. The full anonymised dataset (i.e., all data excluding any team names) is available from the authors on reasonable request.

Ethical approval

This article does not contain any experimental study involving human participants or animals performed by any of the authors.

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